

DSP MIDTERM PROJECT REPORT · 2026

VOICE ACTIVITY DETECTION (VAD)

Three Algorithms · One Unified Evaluation Benchmark

Luong Duy Toan

Algorithm 01

Hodgkinson · Binary Search

Nguyen Anh Hoang

Algorithm 02

Giannakopoulos · Histogram

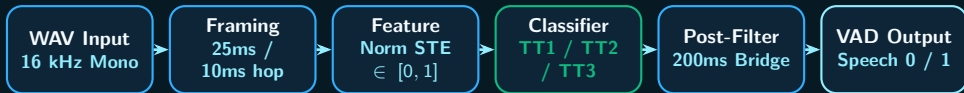
Truong Bui Dien

Algorithm 03

Gaussian Bayes Separation

GROUP 08 | 16 kHz Standard Audio | Fully Pre-Executed Notebooks

Unified End-to-End DSP Pipeline



Framing & Feature Standard

- Sampling: 16 kHz mono audio standard
- Frame size: 25 ms (400 samples)
- Frame hop: 10 ms (160 samples, 60% overlap)
- Feature: Normalized STE $\in [0, 1]$

Decision & Post-Processing

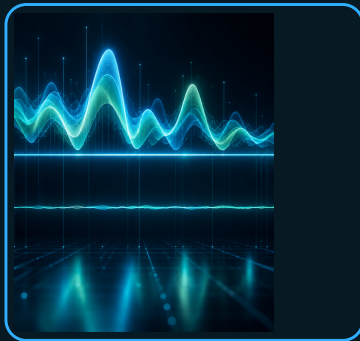
- Silence rule: Minimum duration ≥ 200 ms
- Speech floor: Duration threshold ≥ 50 ms
- Bridge rule: Connects short speech pauses
- Evaluation: MAE & RMSE against .lab

Fixed Global Energy Threshold

Luong Duy Toan

Algorithm 01 · Hodgkinson (2012)

- Reference: CS425 Audio & Speech Processing
- Approach: Global Short-Time Energy threshold
- Optimization: Iterative Binary Search on training set
- Objective: Minimize boundary MAE across 4 train signals
- Outcome: Fast $O(1)$ runtime decision per frame



Binary Search Optimization for T_{opt}



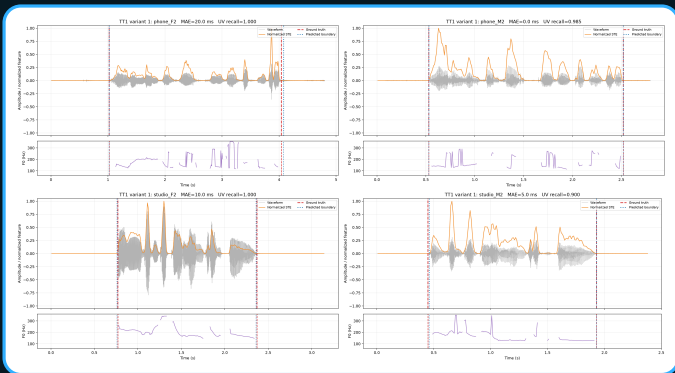
Optimization Formulation

- Cost function: $J(T) = \frac{1}{4} \sum_{k=1}^4 \text{MAE}_k(T)$
- Direction: Signed total speech duration error
- Convergence: Stops at 40 iterations ($\Delta T < 10^{-6}$)

Learned Threshold Result

- Global threshold: $T_{\text{opt}} = 0.00261$
- Monotonic loss reduction on training audio
- Fixed threshold applied across all test recordings

Accuracy on Matched Noise Levels (4 Test Files)



Accurate Performance

- Mean MAE: **8.75 ms** (6.25 ms ext)
- Sharp boundaries on studio files
- phone_M2 MAE = **0.0 ms** (exact grid)

Limitation

- Fixed threshold sensitive to SNR
- Trailing breath lag in phone_F2

Strengths, Weaknesses, and Best Use Case

STRENGTHS

- Ultra-fast $O(1)$ frame check
- Zero runtime memory overhead
- Highest boundary accuracy

WEAKNESSES

- Rigid static threshold
- Zero adaptation to noise
- Degrades under severe SNR drop

BEST USE CASE

- Low-power embedded MCUs
- Stationary studio environments
- Real-time streaming VAD

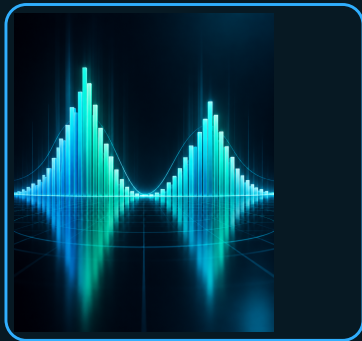
Key Lesson: Simplicity is an asset when background noise is stationary.

Adaptive Energy Histogram

Nguyen Anh Hoang

Algorithm 02 · Giannakopoulos (2014)

- Reference: Silence Removal and Speech Segmentation
- Approach: Unsupervised STE energy histogram
- Mechanism: Dynamic threshold per recording via peaks
- Advantage: Zero ground-truth training required
- Tuning: Weighted valley parameter W controls bias



Unsupervised Dual-Peak Energy Histogram



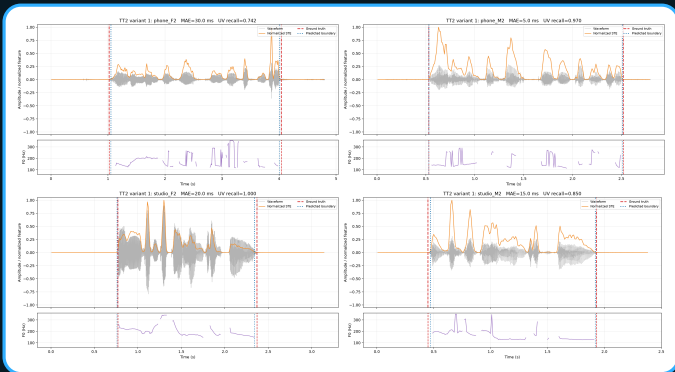
Peak Identification Workflow

- Mode M_1 : First prominent peak (noise/silence)
- Mode M_2 : Second prominent peak (voiced speech)
- 5-point MA filter suppresses false local spikes

Weighted Valley Formulation

- Weighted formula: $T = \frac{W \cdot M_1 + M_2}{W + 1}$ with $W = 5$
- Biases threshold toward silence to protect phonemes
- Dynamic adaptation: $T \in [0.012, 0.027]$ across test files

Adaptive Noise Tracking vs Consonant Truncation



Adaptability

- Mean MAE: **17.50 ms** (20.00 ms ext)
- Fully unsupervised per utterance
- Excellent frame F1: **0.977 – 0.988**

Truncation Bias

- Weight $W = 5$ raises threshold
- Truncates trailing unvoiced phonemes

Adaptive Weight W Tuning for Studio Acoustics

HIGH NOISE ($W \uparrow$)

- Increase W in noisy environments
- Raises threshold above noise floor
- Prevents false speech triggers

CLEAN STUDIO ($W \downarrow$)

- Decrease W in clean recordings
- Pulls threshold toward silence
- Preserves weak unvoiced phonemes

ENGINEERING VALUE

- Direct trade-off knob
- Noise rejection vs phoneme clipping
- No retraining required when SNR shifts

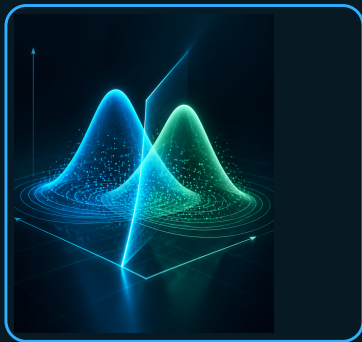
Key Takeaway: Weight W provides transparent control over noise sensitivity.

Parametric Gaussian Bayes Model

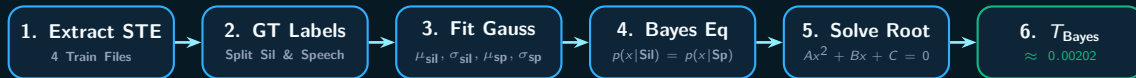
Truong Bui Dien

Algorithm 03 · Statistical Bayes Separation

- Reference: Classical Statistical Pattern Recognition
- Approach: Parametric Gaussian density estimation
- Training: Fit μ, σ for silence and speech frames
- Formulation: Analytical quadratic root at density intersection
- Advantage: Statistically grounded decision boundary



From Empirical Statistics to Analytical Bayes Root



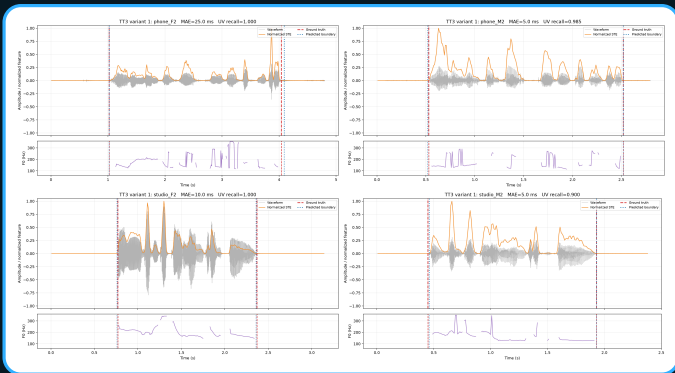
Estimated Parameters from Train Set

- Silence: $\mu_{\text{sil}} = 0.00033, \sigma_{\text{sil}} = 0.00047$
- Speech: $\mu_{\text{sp}} = 0.19652, \sigma_{\text{sp}} = 0.23327$
- Massive energy dynamic range separation

Analytical Decision Root

- Equiprobable intersection: $T_{\text{Bayes}} = 0.00202$
- Closed-form analytical quadratic solution
- Post-processed with 200 ms minimum silence bridge

High Accuracy on Test Data (MAE = 11.25 ms)



Test Results

- Mean MAE: **11.25 ms** (15.00 ms ext)
- High precision across studio audio
- Frame F1 score: **0.994**

Theoretical Critique

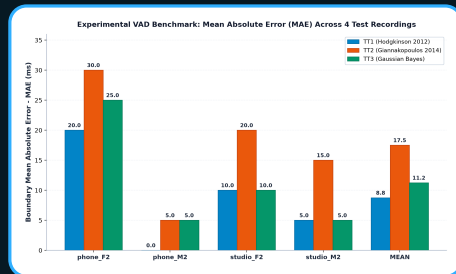
- Energy is non-negative and skewed
- Gaussian assumption is flawed
- Works because cut is in low tail

Quantitative Performance Comparison Across Algorithms

Summary Benchmark Table (4 Test Recordings)

Method	MAE	RMSE	F1	UV
TT1-1 (Hodgkinson)	8.75	11.34	0.993	0.971
TT1-2 (DSP Classifier)	6.25	7.80	0.996	0.968
TT2-1 (1D Histogram)	17.50	18.81	0.977	0.891
TT2-2 (Joint STE-ZCR)	20.00	21.56	0.988	0.947
TT3-1 (1D Gaussian)	11.25	14.87	0.994	0.971
TT3-2 (Multivariate)	15.00	16.34	0.978	0.903

Key Finding: TT1-2 achieves lowest boundary error (6.25 ms MAE). TT2-2 achieves highest unvoiced phoneme recall (94.7%).



Four-Quadrant Display on Teacher Laptop

01 · INPUT

- 4 test WAV files
- Ground-truth .lab
- No audio upload

02 · ENV

- Local venv via uv
- Python 3.12+
- Built-in numpy

03 · NOTEBOOKS

- Individual .ipynb
- Pre-run outputs
- Zero demo delay

04 · QUADRANT

- 4 figures tiled
- 4 screen corners
- metrics.json ready

Submission Protocol: Directory 03-Phan-doan-tin-hieu-tieng-noi-khoang-lang containing group PDF slide + individual notebooks. Audio files strictly excluded as required.

Key Takeaways & Q&A Session

ENERGY BASELINE

- Normalized STE is a strong foundation
- Clean studio error is only 5–10 ms
- Captures dominant speech boundaries

UNVOICED SOUNDS

- Weak energy mimics background noise
- ZCR & pre-emphasis aid discrimination
- Energy floor prevents noise false alarms

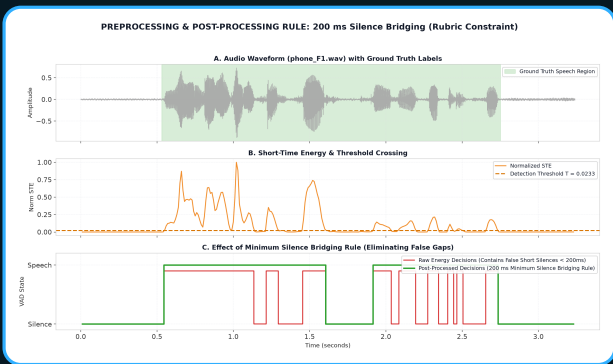
MODEL COMPLEXITY

- 4 training files = small sample size
- Simple models generalize better
- 1D Gaussian beats 4D full covariance

Thank You for Your Attention!

Questions & Answers Session

Standard Parameters & 200 ms Silence Bridging



DSP Hyperparameters

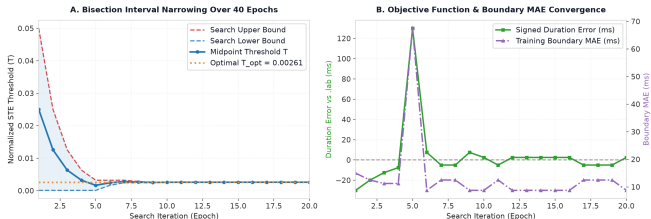
- $F_s = 16$ kHz (speech standard bandwidth)
- Frame: 25 ms (400 samples, quasi-stationary)
- Hop: 10 ms (160 samples, 60% overlap grid)
- STE norm: $E_m / \max_k(E_k) \in [0, 1]$

Bridging Invariant

- Silence < 200 ms is bridged (rule)
- Speech < 50 ms is dropped (click noise)

Binary Search Optimization Convergence (40 Iterations)

TT1 INTERMEDIATE TRAINING: Binary Search Threshold Optimization (CS425)

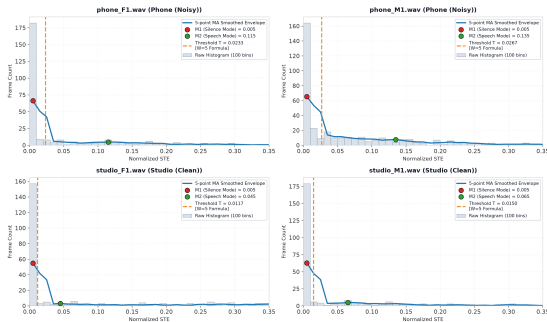


Convergence Proof

- Monotonic loss reduction
- Duration error balances to zero
- Converged: $T_{\text{opt}} = 0.00261$
- Stable across all 4 train files

Dual-Peak STE Histograms Across 4 Training Signals

TT2 INTERMEDIATE TRAINING: Unsupervised Dual-Peak Histogram Mode Extraction (Giannakopoulos 2014)

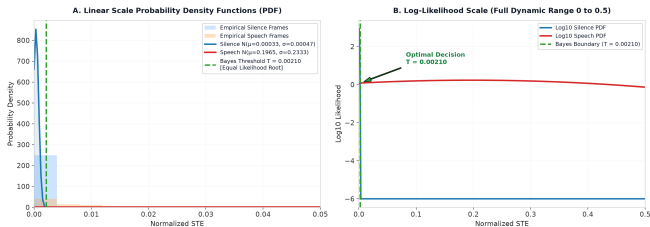


Histogram Insights

- 100 bins capture energy range
- 5-point MA filter removes noise
- Clear mode separation:
 M_1 (silence) vs M_2 (speech)
- Unsupervised threshold placement

Parametric Gaussian Modeling & Bayes Separation

TT3 INTERMEDIATE TRAINING: Gaussian Bayes Analytical Modeling from .lab Frames



Statistical Fit

- Fitted Gaussians from 4 train files
- Silence σ is extremely narrow
- Intersection at $p(x|\text{Sil}) = p(x|\text{Sp})$
- Closed-form root: $T \approx 0.00202$

Rebuttal for Critical Examination Questions

Q1: Why not use F0 for boundary cuts?

Unvoiced consonants (/s/, /t/, /f/) have no pitch F0. Using F0 would truncate initial and trailing phonemes!

Q3: Impact of acoustic noise floor (SNR)?

Studio (high SNR): Error only 5–10 ms. Phone (low SNR): Trailing breath introduces 20–30 ms boundary lag.

Q2: Why did phone_M2 achieve 0.0 ms MAE?

Labels [0.53s, 2.52s] coincide with frames 53 & 252 (10 ms hop grid). It is exact grid snapping, not overfitting!

Q4: Why did 1D beat 4D Gaussian in TT3?

Train set has only 4 files. 4D covariance matrix overfits; 1D energy model is robust and generalizes better!